Precision Z Stage Suitable for Vacuum

Compact Integration with Linear and Rotation Stages

L-310 V6



- Suitable for vacuum up to 10^{-6} hPa
- Travel range 26 mm (1")
- High-resolution encoder
- Stepper motor
- Zero-play ball screw
- Load capacity to 5.5 kg

Precision-class Z stage

High guide accuracy and stiffness due to ball screws and crossed roller guides. Stress-relieved aluminum base for high stability. Noncontact limit and reference switches (Hall effect). Reference switch with direction sensing in the middle of the travel range. Integrated linear encoder, installed centrically.

Applications and use cases

- Vacuum chambers
- Research
- Beamline instrumentation
- Microscopy
- Electronics assembly and inspection

Technology

Drive type: 2-phase stepper motor for high torque even at low velocities and high resolution.

Position measuring: Noncontact optical linear encoders measure the position directly at the platform with the greatest accuracy. Nonlinearity, mechanical play or elastic deformation have no influence on the measurement.

Reduced outgassing especially for use in vacuum environments: In particular, the guide and the spindle are lubricated using a very low-outgassing substance. This allows a bakeout temperature of up to 80 °C. This and other modifications to the PI V6 vacuum class make it possible to use this positioner at up to 10^{-6} hPa.

Technical data

Motion	Unit	Tolerance	L-310.021211V6
Active axes			Z
Travel range in Z	mm		26
Maximum velocity in Z, unloaded	mm/s		7.5
Straightness error in X	µm	typ.	±3
Straightness error in Y	µm	typ.	±3
Angular error around X	µrad	typ.	±150
Angular error around Y	µrad	typ.	±150

Positioning	Unit	Tolerance	L-310.021211V6
Minimum incremental motion	µm	typ.	0.05
Unidirectional repeatability	µm	typ.	±0.05
Bidirectional repeatability	µm	typ.	0.4
Reference switch			Hall effect, N/C contact, 5 V, NPN
Reference switch repeatability	µm		2
Limit switch			Hall effect, N/C contact, 5 V, NPN
Integrated sensor			Incremental linear encoder
Sensor signal			Sin/cos, 1 V peak-to-peak
Sensor signal period	µm		20

Drive properties	Unit	Tolerance	L-310.021211V6
Drive type			2-phase stepper motor
Nominal voltage	V		24
Motor resolution	Full steps/ rev.		200
Drive force	N	typ.	55

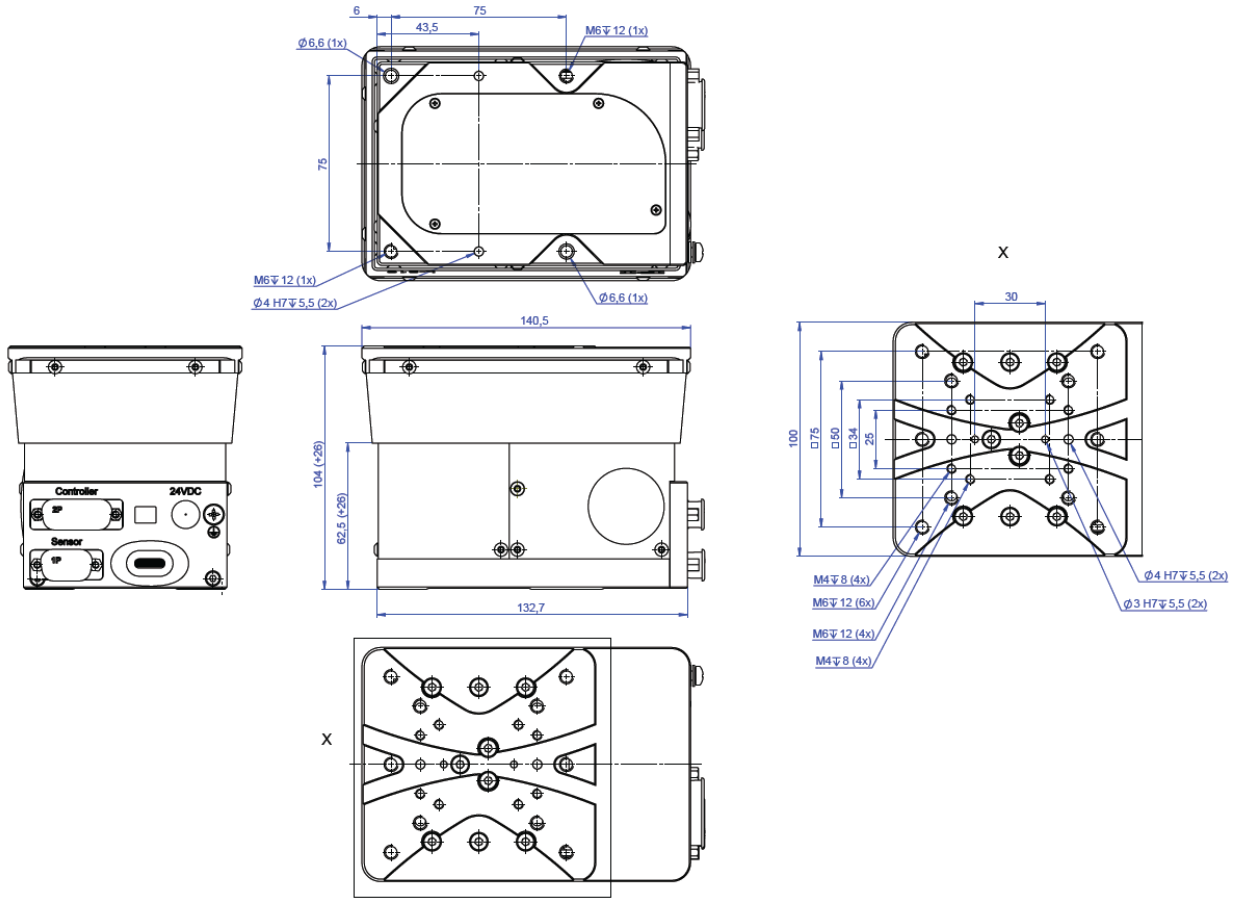
Mechanical properties	Unit	Tolerance	L-310.021211V6
Permissible push force in X	N	max.	100
Permissible push force in Y	N	max.	50
Permissible push force in Z	N	max.	55
Permissible torque in θX	Nm	max.	40

Mechanical properties	Unit	Toler-ance	L-310.021211V6
Permissible torque in θY	Nm	max.	80
Permissible torque in θZ	Nm	max.	80
Holding force in Z, passive	N		50
Moved mass in Z, unloaded	kg		0.9
Drive screw type			Ball screw
Spindle pitch	mm		1
Guide			Crossed roller guide
Overall mass	kg		2.6
Material			Anodized aluminum

Miscellaneous	Unit		L-310.021211V6
Operating temperature range	°C		5 to 40
Vacuum class	hPa		10
Connector			HD D-sub 26 (m)
Sensor connector			D-sub 9 (m)
Recommended controllers/drivers			C-663.12, C-885 with C-663.12C885 ACS modular controller

At PI, technical data is specified at 22 ±3 °C. Unless otherwise stated, the values are for unloaded conditions. Some properties are interdependent. The designation "typ." indicates a statistical average for a property; it does not indicate a guaranteed value for every product supplied. During the final inspection of a product, only selected properties are analyzed, not all. Please note that some product characteristics may deteriorate with increasing operating time.

Drawings and images



L-310.021211V6, dimensions in mm

Ordering information

L-310.021211V6

Precision Z stage suitable for vacuum; 2-phase stepper motor; 26 mm travel range; 55 N load capacity; 7.5 mm/s maximum velocity; ball screw; incremental linear encoder, 20 µm sensor signal period, sin/cos, 1 V peak-peak; vacuum-compatible to 10^{-6} hPa

Accessories

6100V075

AB-65 L mounting bracket for vertical mounting of linear and rotation stages; aluminum, blank; vacuum-compatible to 10^{-9} hPa

C-815.VF

Vacuum feedthrough, D-sub 15 (m/f), suitable for high vacuum (to 10^{-6} hPa)

C-815.VF26

Vacuum feedthrough, HD D-sub 26 (m/f), suitable for high vacuum (to 10^{-6} hPa)

C-815.VF9

Vacuum feedthrough, D-sub 9 (m/f), suitable for high vacuum (to 10^{-6} hPa)

C-815.VFU1

Vacuum feedthrough, CF40 flange, 1 × D-sub 15 (m/m), suitable for high vacuum (to 10^{-7} hPa) and ultra-high vacuum (to 10^{-9} hPa)

C-815.VFU3

Vacuum feedthrough, CF63 flange, 3 × D-sub 15 (m/m), suitable for high vacuum (to 10^{-7} hPa) and ultra-high vacuum (to 10^{-9} hPa)

C-815.VFU6

Vacuum feedthrough, CF100 flange, 6 × D-sub 15 (m/m), suitable for high vacuum (to 10^{-7} hPa) and ultra-high vacuum (to 10^{-9} hPa)

C-815.VFU15

Vacuum feedthrough, CF160 flange, 15 × D-sub 15 (m/m), suitable for high vacuum (to 10^{-7} hPa) and ultrahigh vacuum (to 10^{-9} hPa)

C-815.VFU11

Vacuum feedthrough, CF63 flange, 1 × D-sub 9 (m/m) and 1 x D-sub 15 (m/m), suitable for high vacuum (to 10^{-7} hPa) and ultrahigh vacuum (to 10^{-9} hPa)

C-815.VFU33

Vacuum feedthrough, CF100 flange, 3 × D-sub 9 (m/m) and 3 x D-sub 15 (m/m), suitable for high vacuum (to 10^{-7} hPa) and ultrahigh vacuum (to 10^{-9} hPa)

C-815.VFU66

Vacuum feedthrough, CF160 flange, 6 × D-sub 9 (m/m) and 6 x D-sub 15 (m/m), suitable for high vacuum (to 10^{-7} hPa) and ultrahigh vacuum (to 10^{-9} hPa)

6100V075

AB-65 L mounting bracket, vacuum-compatible to 10^{-9} hPa

L-500.AV3-V6

Mounting bracket for vertical mounting of positioners up to 100 mm in width and up to 300 mm in length; vacuum-compatible to 10^{-6} hPa

L-500.AV3-V9

Mounting bracket for vertical mounting of positioners up to 100 mm in width and up to 300 mm in length; vacuum-compatible to 10^{-9} hPa